

using gates is given by:

$$f = \frac{1}{2.2(RC)}$$

You can vary R5 and C3 values using variable resistors and capacitors to fine-tune the frequency.

Assemble the circuit on a general-purpose PCB and enclose in a suitable small cabinet.

Terminate all the four terminals of the coils on the PCB base for connecting the coils to the main circuit (refer Fig. 3). The gel-pen refill should be sufficiently rugged. Secure it firmly inside the bobbin using non-magnetic, non-conducting materials. The ferrite rods too must be sufficiently secured in their positions using a synthetic enamel. Even a slight unintentional displacement can upset the balance

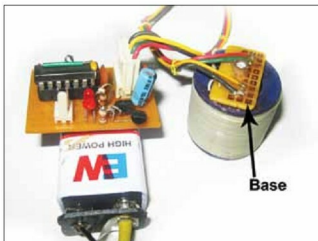


Fig. 3: Author's prototype

of the resonator drastically. So use of a screw-type ferrite rod is recommended.

For detector coil assembly, first

insert one of the ferrite rods in the front end of the detector such that it is just inside the gel refill tube. Now insert the second ferrite rod in the other end of the tube. At this point, LED1 should glow brightly. Push the ferrite rod very slowly inside the tube while watching LED1. As soon as LED1 goes off, stop pushing the

rod, mark the position of the rod and fix it in the tube using glue. Now the detector is well adjusted and ready for use. ●

USB-POWERED PC LAMP

■ D. MOHAN KUMAR

This white LED lamp lets you use your PC or laptop at night without disturbing others' sleep. It produces a soft white light just enough to see the keyboard in darkness as well as when the ambient light is poor during daytime. The circuit is powered by regulated 5V DC available from the USB socket of the PC.

LDRI acts as a light-dependent switch to turn on the lamp (LED2 through LED7) when the ambient light in the room drops below the preset level. Transistors T1 and T2 (BC547) are used to switch on the lamp. The base of transistor T1 is connected to the voltage divider comprising LDRI and preset VR1. When light in the room is sufficient, the resistance of LDRI is low.

This results in a high voltage at the base of T1, driving it into saturation. When transistor T1 conducts, transistor T2 is cut off. This disconnects the power supply to all the white LEDs (LED2 through LED7). LED1 (green LED) glows as it is forward-biased, indicating the standby mode.

When it is dark, or the ambient light in the room is less than the predetermined level set by VR1, transistor T1 is cut off and T2 conducts. All

the white LEDs glow with sufficient brightness as these are connected to the power supply through series dropper resistors R2 through R7. These resistors are used to limit the current through white LEDs to a safe level.

White LEDs are arranged in parallel as each white LED requires a minimum of 2V. Preset VR2 is used to set the base voltage of transistor T2. Adjust preset VR2 until the white LEDs turn off in the preset light intensity level during the day.



Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Power to the circuit is obtained from the USB socket using a USB cable. You can use an old USB cable for the purpose. Cut the ends of the USB cable

